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ORIGINAL PAPER

Nonverbal Behaviour in Football: A 3D Pose-Based Analysis of Player Expressions and Observer Perceptions

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Abstract

Nonverbal behaviour (NVB) is a critical component of human communication, particularly in high-stakes, naturalistic settings such as sport. Despite advances in computational analysis, the extent to which objectively computable NVB features reflect subjective human judgments remains unclear. In this study, we present the inaugural investigation into the correspondence between computable NVB indicators and observer perceptions in the context of elite football. The utilisation of 3D pose data from a FIFA World Cup match enabled the extraction of volumetric, angular and kinematic features from players' body movements during match events. Parallelly, human participants rated short video segments of these events on multiple NVB dimensions. Statistical analyses revealed significant differences in objective computable NVB indicators across play contexts (scoring, conceding, regular play), with players on the scoring team exhibiting more expansive and dynamic body language. Subjective ratings closely mirrored these differences, and repeated measures correlation analyses demonstrated strong associations between objectively computable NVB features and human judgments. These findings may provide a scientific basis for the use of 3D pose-based metrics for scalable, automated NVB analysis and may highlight their ecological validity in capturing expressive behaviours meaningful to human observers. The study has the potential to contribute to the advancement of methodological approaches for NVB research.

Keywords Nonverbal behaviour · Body language · 3D pose estimation · Computational analysis · Person perception

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Introduction

Nonverbal behaviour (NVB) is a fundamental aspect of human communication, conveying emotions, intentions, and attitudes through modalities such as facial expressions, posture, gestures, and movement (Wiener et al., 1972). As Watzlawick's assertion that "one cannot not communicate" highlights, decades of research on nonverbal communication have demonstrated that it is impossible for people to refrain from communicating nonverbally. This is because every cue—or even the absence of cues—is likely to be interpreted by others (Watzlawick et al., 2011). Notwithstanding the well-established importance of NVB, systematic research examining how objectively computable features of body language correspond to the subjective impressions formed by observers remains limited, particularly in naturalistic and high-stakes contexts such as sport. The present study addresses this gap for the first time by examining the relationship between objectively computable NVB features and human judgments, using 3D pose data obtained from actual football matches.

Sport offers a unique and valuable context for studying NVB, as it is one of the few social environments that is extensively documented and provides a rich array of naturally occurring, intense nonverbal expressions (Furley, 2019). Sports commentators often reference athletes' body language as an indicator of psychological state or momentum, highlighting the perceived importance of such behaviours (Furley, 2021). Empirical research has shown that athletes' body language is systematically linked to key psychological constructs, such as confidence (Hays et al., 2009), which are widely recognised as critical for successful performance outcomes (Furley, 2021). This underscores the relevance of NVB as a meaningful and interpretable source of information in competitive sport.

Among studies of NVB, the human face is widely regarded as the most important channel for conveying nonverbal information (Matsumoto et al., 2013). Major advances in affective computing and emotion research have been achieved through manual coding systems such as the Facial Action Coding System (FACS; Ekman & Friesen, 1978) and automated facial expression recognition tools like Noldus FaceReader 7. While facial expression research has progressed rapidly, methodological development in the study of body movement has lagged behind. This is largely due to the greater complexity and degrees of freedom in whole-body movement compared to facial expressions (Furley & Roth, 2021). To address these challenges, automated motion capture systems and descriptive coding frameworks—such as the Body Action and Posture Coding System (BAP; Dael et al., 2012) and the Nonverbal Behaviour Coding System for Soccer Penalties (NBCSP; Furley & Roth, 2021) have been introduced. However, motion capture systems are typically limited to highly controlled laboratory settings (Bente et al., 2008), and descriptive coding systems like BAP and NBCSP require extensive training and are time-consuming to implement and analyse (Furley & Roth, 2021).

In recent years, there has been a proliferation of technological advances that offer a promising alternative to traditional methods. In particular, advancements in the fields of computer vision and deep learning have led to the development of novel tools that have the potential to overcome some of the limitations associated with both manual coding systems and motion capture technologies. These technologies facilitate the reconstruction of human posture and movement from video footage, generating three-dimensional meshes and skeletal representations through the identification of key joints and the analysis of pose dynamics (Jiang et al., 2024; Li et al., 2022; Park et al., 2023). The identification of key body joints and key points facilitates a detailed analysis of body movements. Such methods have the potential to enable fine-grained, scalable and less resource-intensive analysis of NVB, particularly in naturalistic contexts outside the laboratory.

However, there has been a paucity of systematic research into which objective computable indicators derived from 3D pose data can reliably quantify aspects of body language. Furthermore, the extent to which these objective computable indicators correspond to the subjective impressions and evaluations formed by human observers remains unclear.

To address these gaps, the present study adopted a two-phase approach. First, objective computable measures of players' bodily expressions were extracted from 3D pose estimation data. Next, participants viewed corresponding video clips and provided subjective ratings of the players' psychological states. Objective computable measures were statistically compared across different play contexts (e.g., scoring vs. conceding goals) to identify indicators that may be able to quantitatively represent aspects of NVB. The relationship between subjective ratings and objective computable indicators was then examined to determine which features best reflected observers' perceptions of NVB. The present methodology draws inspiration from Brunswik's lens model (Brunswik, 1956), a framework for analysing how nonverbal cues relate to a criterion variable (i.e. cue validity) and how observers use these cues to form judgments (i.e. cue utilisation).

To the best of our knowledge, this is the first study to investigate the relationship between human judgments of NVB and objective computable features derived from 3D pose estimation data obtained from actual football matches. It is anticipated that the findings resulting from this approach will contribute to the development of scalable, automated methodologies for analysing whole-body nonverbal communication.

Methods

Data

For the purposes of this research, a dataset corresponding to a single 2022 FIFA World Cup match was analysed; namely, the Round of 16 game between the USA and the Netherlands. This subset includes 2D positional data for each player and the ball (sampled at 50 Hz), as well as 3D coordinates for anatomical keypoints (also at 50 Hz). The keypoints cover major anatomical landmarks—such as ankles, knees, hips, shoulders, elbows, wrists, fingers, toes, ears, eyes, heels, and nose—along with central reference points including the mid-hip and neck.

Data Processing

Selection of Scenes

The scenes selected for analysis were identified on the basis of pivotal events within the match. The analysis focused on eight key moments within the match: First half kick-off, first half end, second half kick-off, second half end, first goal, second goal, third goal and fourth goal. For each of the kick-off and end of half events, a 20-s segment was extracted, consisting of 10 s before and 10 s after the respective timestamp. In contrast, for each goal event, only the 10 s immediately following the goal were extracted. The four goals in the USA vs. Netherlands match occurred as follows: Netherlands 1–0 (10′), Netherlands 2–0 (45 + 1′), USA 1–2 (76′), and Netherlands 3–1 (81′).

The 10-s video windows were selected on the basis of research demonstrating that brief "thin slices" of several seconds are sufficient for reliable evaluations of athletes' NVB (Furley et al., 2021, 2025). Key moments during the match that were likely to prompt pronounced NVB from the players were selected, such as situations involving goals. The rationale for the selection of scenes was to contrast highly significant match events (i.e., goals) with more neutral events (kick-off before match and half time). Arguably such significant match events will elicit stronger internal states like emotions and coinciding nonverbal behaviors as more neutral match situations. Previous research has supported this reasoning as goals in team sports events have been shown to be associated with intense emotional and behavioural responses (Moesch et al., 2014; Moll et al., 2010). The reasoning for scene selection is further backed up by appraisal approaches to emotion (e.g., Scherer, 2013) or self-regulation theory (Carver & Scheier, 1990) that both emphasize the centrality of goal-conduciveness (i.e., the extent to which an event facilitates or obstructs one's goals) as a key appraisal dimension for affective processes (see Yeo & Ong, 2024 for a recent meta-analysis). Applied to the present context, a scored goal is likely to be appraised by the team as goal-conducive (of winning the game), thereby eliciting positive affective reactions and corresponding NVB, whereas a conceded goal is appraised as goal-obstructive (of winning the game), eliciting negative affective responses and corresponding NVB. A thorough delineation of the methodology employed for the determination of event times can be found in the Supplemental Information.

Generation of 3D Pose-Based NVB Video Sequences

The NVB analysis, which was based on 3D pose estimation, included all players who were present on the pitch during each scene that was selected. For each participant, 3D coordinates of anatomical key points (X, Y, Z) were utilised to generate frame-by-frame 3D visualisations, which were compiled into scene-specific video sequences (approximately 500 frames, 500 × 500 pixels, approximately 10 s per sequence). The total number of video sequences thus generated was 248, with each sequence corresponding to a specific player and scene. Figure 1 presents an example of a three-dimensional skeletal representation from the dataset.

Selection of Objective Computable NVB Features

The selection of objective computable NVB features focused on nonverbal cues that are potentially diagnostic of underlying psychological states and that observers are likely to use when forming judgments. Specifically, the following measures were selected for analysis: total body bounding box volume, upper body convex hull volume, body bend angle, face tilt angle, and instantaneous speed.

Volumetric features possess the capacity to capture the spatial extent of the body and are indicative of the degree of expansiveness or contraction in posture. As indicated by previous research, expansive postures are indicative of dominance and confidence, whilst contracted postures are associated with submission or low power (Carney et al., 2005, 2010; Tracy &

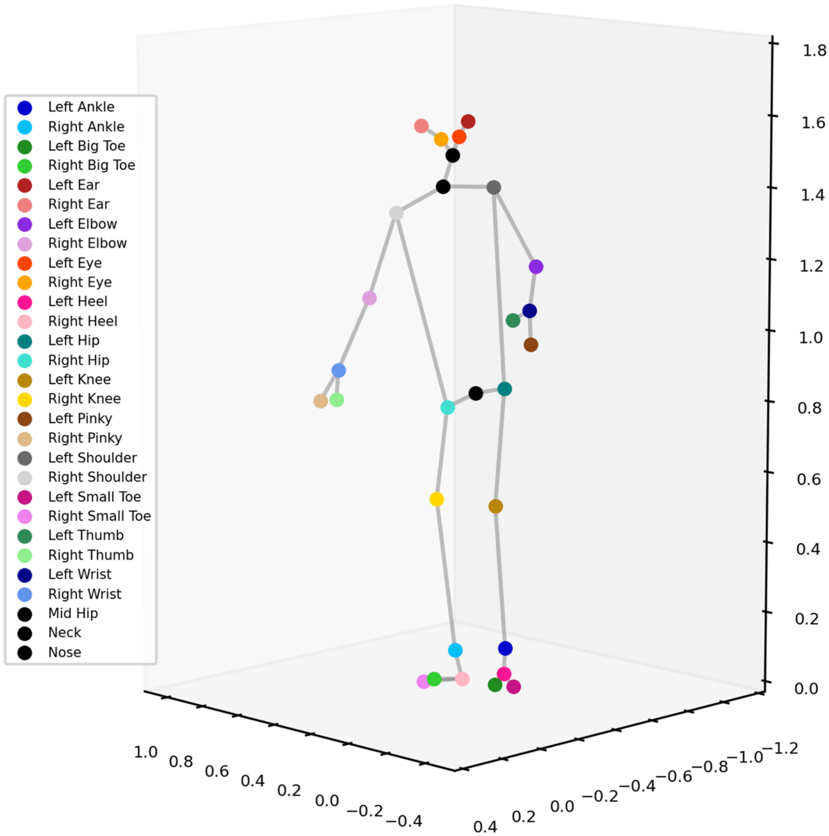

Fig. 1 Example of a 3D skeletal representation of a single player at one point in time. Each marker represents a specific anatomical joint, colour coded by body part, and the grey lines indicate the connections between adjacent joints

Matsumoto, 2008). Angular features have been demonstrated as a means of quantifying postural cues. Emotions associated with failure, such as shame or embarrassment, are typified by a downward gaze and rigid, slouching, forward-leaning posture (Keltner, 1995; Tracy & Matsumoto, 2008). Instantaneous speed was included as movement dynamics, such as walking speed, systematically vary with emotional states and influence observers' perceptions (Gross et al., 2012; Montepare et al., 1987).

Quantification of NVB Features from 2D Positional Data and 3D Pose Estimation Data

In order to quantify the NVB of players in video sequences, a set of features was calculated on the basis of 2D positional data and 3D pose estimation data. To quantitatively assess the spatial extent of the body in three dimensions, two volume-based measures were computed from the joint coordinate data. These corresponded to (1) total body volume and (2) upper body volume. The initial step in the process was to calculate the three-dimensional bounding box that enclosed all of the tracked joints in each frame, in order to approximate total body volume. This was achieved by determining the minimum and maximum values of the joint coordinates along each of the three axes (x, y and z), thereby defining an axis-aligned rectangular prism that fully encompasses the body at that particular moment in time (Fig. 1). The volumes of the bounding boxes were calculated using the following equations:

$$\mathrm{Volume} = \mathrm{w} \times \mathrm{h} \times \mathrm{d} \tag{1}$$

where w, h and d are the width, height and depth of the bounding box respectively.

Secondly, in order to obtain a measure of the upper body that is more anatomically focused, the volume of the convex hull enclosing a selected subset of upper-body joints was calculated. Specifically, the 3D coordinates of the left and right shoulders, elbows, wrists, hips, thumbs and little fingers, in addition to the mid-hip and neck landmarks, were utilised. The volume enclosed by the convex hull was computed using the ConvexHull class from the scipy.spatial module (Virtanen et al., 2020). The convex hull is defined as the smallest convex shape that can enclose a set of points in three-dimensional space (Fig. 2).

In order to quantify postural characteristics from 3D keypoint data, two types of angular feature were computed: body bend angle and face tilt angle. The 3D coordinates of the mid-hip, neck, and nose keypoints in each frame of the dataset were utilised to derive these. Initially, the body bend angle was calculated as the angle between the vector from the mid-hip to the neck and the vertical Z-axis. The purpose of this metric is to quantify the extent to which the upper body bends forward or backward. The face tilt angle was then defined as the angle between the vector from the neck to the nose and the vector from the hip to the neck, taken through the midpoint of the body. This angle thus indicates the relative orientation of the head and torso, thereby demonstrating whether the face is aligned or tilted with respect to the body's axis. The angle between the two 3D vectors was calculated for each frame using the formula for the dot product (Fig. 3). Each vector was first normalised to unit length. The angle θ in radians between two vectors $\vec{v}_1$ and $\vec{v}_2$ was calculated as follows:

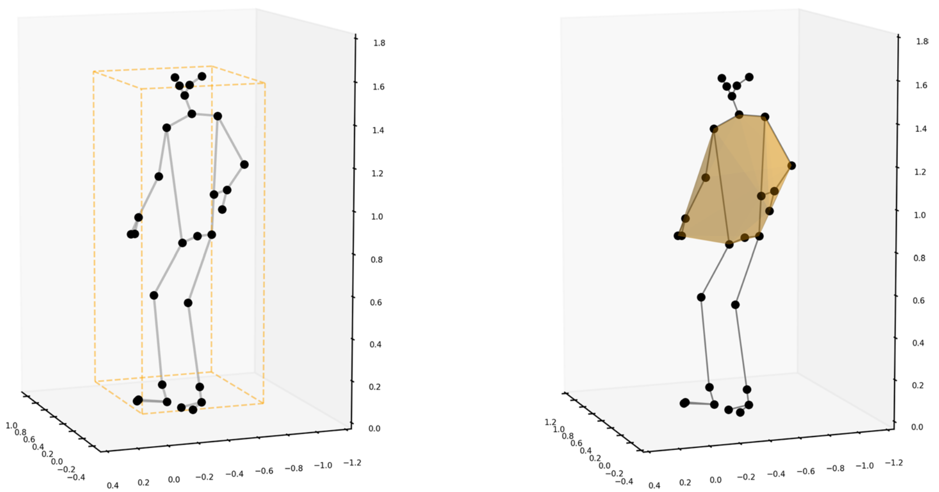

Fig. 2 Illustration of the calculation of total body volume and upper body volume from 3D keypoint data. The total body volume is represented by the axis-aligned bounding box (left), while the upper body volume is represented by the convex hull (right) enclosing selected upper-body joints

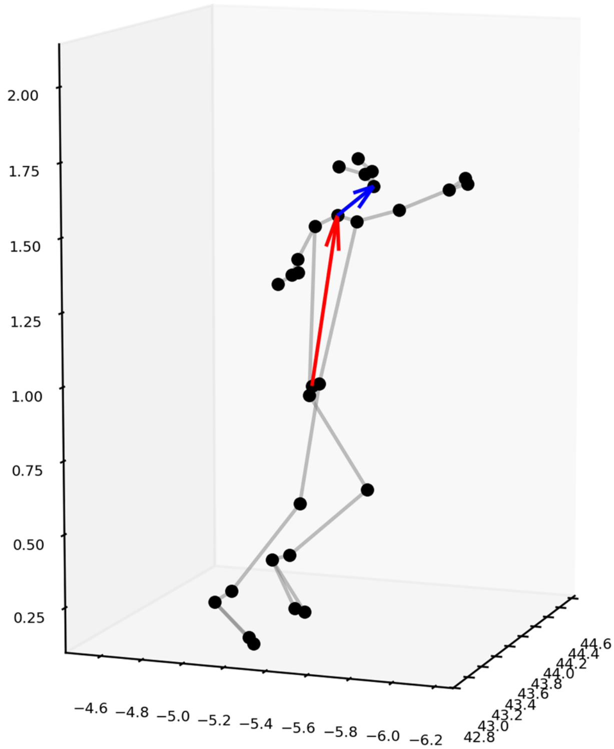

Fig. 3 Illustration of the calculation of body bend angle and face tilt angle from 3D keypoint data. The red arrow represents the vector from the centre of the hips to the neck, used to quantify the tilt of the trunk relative to the vertical axis. The blue arrow represents the vector from the neck to the nose, used to quantify the tilt of the head relative to the trunk

$$\theta = \cos^{-1}\left(\frac{\vec{v}_1 \cdot \vec{v}_2}{||\vec{v}_1|| \cdot ||\vec{v}_2||}\right) = \cos^{-1}\left(\frac{x_1x_2+y_1y_2+z_1z_2}{\sqrt{x_1^2+y_1^2+z_1^2}\cdot\sqrt{x_2^2+y_2^2+z_2^2}}\right) \tag{2}$$

The resulting angle in radians was then converted to degrees:

$$\theta^\circ = \theta \cdot \frac{180}{\pi} \tag{3}$$

In addition to the 3D pose features, a measure derived from the 2D positional data of the players was utilised. Specifically, the instantaneous speed, denoted by v_t, of each player at time t was calculated. The distance traversed between successive frames was utilised, with the resultant figure subsequently being divided by the temporal interval between said frames. The instantaneous speed was computed using the following formula:

$$v_t = \frac{d_t}{\Delta t} \tag{4}$$

Processing of Tactical View Video Segments

In order to facilitate the comparison of human judgments with the objective computable NVB measures derived from 3D pose estimation data, a set of short video segments was extracted from the Tactical View footage of the match. The segments under consideration were selected to correspond to the same match events as those utilised in computing the 3D pose-based NVB features. The complete pipeline for processing short video segments is shown in Fig. 4. To enhance the visual quality of these video segments, a super-resolution technique was applied using the Fast Super-Resolution Convolutional Neural Network (FSRCNN) model (Dong et al., 2016). After super-resolution, individual players were detected using a pre-trained object detection model trained on the Roboflow football player dataset (Roboflow, 2025), and tracked across frames with the DeepSORT algorithm (Wojke et al., 2017). Video segments were manually reviewed to ensure reliable player detection, and only those with successful detection were included. The final dataset comprised 248 cropped video segments focusing on individual players, in mp4 format at 25 frames per second (mean duration: 8 s).

Participants

A total of 332 participants were recruited for an online study on perceptions of NVB in sport. After obtaining informed consent, 305 participants (mean age = 26 years, SD = 8; 221 males, 83 females, 1 undisclosed) were included in the final analysis. Twenty-seven participants were excluded due to incomplete responses or technical issues.

Procedure

The experiment was conducted using a custom web-based application developed in Python (version 3.12.8; Python Core Team, 2025) with the Streamlit framework. This platform pro-

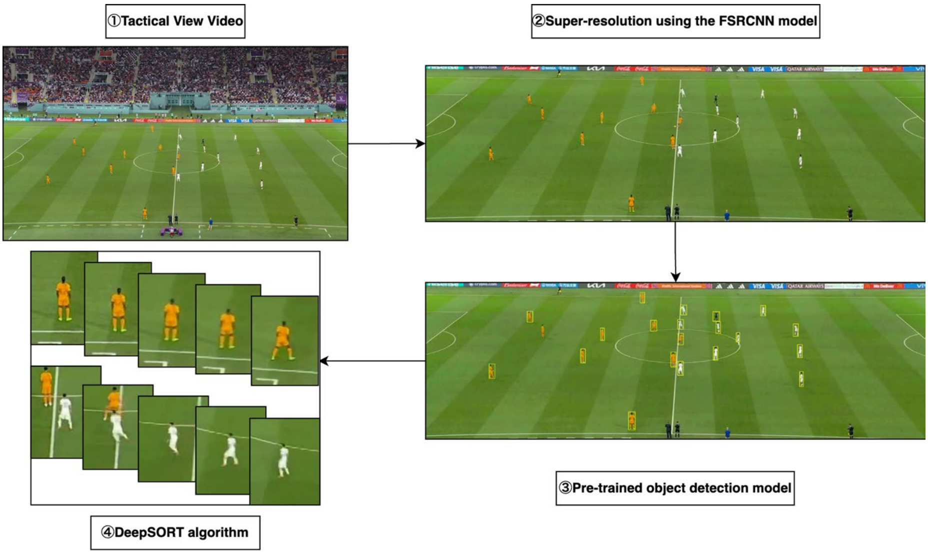

Fig. 4 Workflow for player detection, isolation and tracking in tactical football video. Short video segments corresponding to critical match moments were extracted from the tactical view footage (①). Super-resolution processing using the FSRCNN model was applied to improve video quality (②). A pre-trained object detection model optimised for footballers and trained on the Roboflow dataset was then used for player detection (③). The DeepSORT algorithm was then used to track individual players across successive frames (④)

vided a standardised and user-friendly interface for presenting video stimuli and collecting behavioural ratings, ensuring consistent experimental conditions for all participants. Upon accessing the platform, participants received an introduction to the study and completed an informed consent form. They then answered demographic questions about age, gender, and football experience.

Participants were randomly assigned to one of two video conditions: (1) 3D pose estimation videos (N = 154, age 26 ± 9), showing scene-specific sequences reconstructed from the dataset, or (2) tactical video footage (N = 151, age 26 ± 8), consisting of short segments from the Tactical View broadcast with individual players isolated. The rationale for having two video conditions was rather explorative and enabled examination of whether the biological motion information relating to affective NVB during a football match leads to similar inferences compared to viewing video footage (Marsh et al., 2006). Each participant viewed 50 video clips in random order. For each clip, participants rated the player's NVB using four continuous visual analogue scales (0–100, midpoint 50): (1) negative–positive, (2) submissive–dominant, (3) ashamed–proud, and (4) not confident–confident. In accordance with the findings, which demonstrated the capacity to evaluate athletes' NVB through "thin slices" on dimensions such as dominance, pride, and confidence (Furley et al., 2023), these scales were adopted. The interface allowed participants to pause, rewind, or replay videos to ensure accurate assessment.

Statistical Analysis

All statistical analyses were performed using R (version 4.4.2; R Core Team, 2024) and Python (version 3.12.8; Python Core Team, 2025). Objective computable NVB measures were extracted from 3D pose estimation and 2D positional data, and analysed in relation to play context. Play contexts were classified into three categories: "Scoring Team" (players on the team that scored a goal), "Conceding Team" (players on the team that conceded a goal), and "Regular Play Phase". For a period of 10 s following the commencement of each half, the player of both teams was designated as being in the "Regular Play Phase". To account for individual differences in body size, total body bounding box volume and upper body convex hull volume were normalised by dividing by the cube of each player's height, with height data obtained from the FBref database. For each player and event, mean values of the objective computable measures were calculated across all frames within the corresponding video segment.

The effect of play context on objective computable NVB measures was analysed using linear mixed-effects models (LMMs) with the lme4 package (Bates et al., 2015). Each model included play context as a fixed effect and player ID as a random intercept to account for individual differences. Statistical significance of fixed effects was assessed using Type III F-tests with Satterthwaite's approximation for degrees of freedom (Satterthwaite, 1946). Effect sizes were quantified using omega-squared (ω^2), interpreted as small (ω^2 = 0.01), medium (ω^2 = 0.06), or large (ω^2 = 0.14) (Field, 2018). Model assumptions were checked by examining Q-Q plots for normality and scatter plots of residuals for homoscedasticity and linearity. Where necessary, variables were log10 or square root transformed to meet these assumptions, and Cook's distance was used to identify influential outliers (see Supplementary material for details). When significant fixed effects were found, post-hoc pairwise comparisons were performed using Sheffe's method with the emmeans package (Lenth,

2025) to estimate marginal means and test differences between play contexts. Effect sizes for pairwise comparisons were reported as Cohen's d, interpreted as small (0.2), medium (0.5), or large (0.8) (Cohen, 2013). Statistical significance was set at $p<0.05$ with 95% confidence intervals.

To account for the repeated presentation of identical video segments to multiple participants, the mean value of subjective ratings was calculated for each player, event, and video type. This aggregation ensured that subsequent analyses reflected the central tendency of observer ratings for each unique player-event combination (see Furley et al., 2025). To examine the relationships among the subjective rating dimensions, repeated measures correlation (r_{rm}) analyses were conducted for each of the four rating dimensions within each video type (i.e., eight observed variables in total) using the rmcorr package (Bakdash & Marusich, 2024). This approach was chosen because the same player appeared in different scenes, resulting in non-independence of observations within each player. Repeated measures correlation appropriately accounts for this within-subject structure by estimating the common association between variables across individuals while controlling for inter-individual variability (Bakdash & Marusich, 2017; Bland & Altman, 1995).

In order to provide a more parsimonious summary of the subjective ratings and establish a shared evaluative axis that is not contingent upon the type of video, a single principal component analysis (PCA) was conducted on the four rating dimensions pooled across the two video types (3D pose and tactical) using the PCA class from the sklearn.decomposition module (Pedregosa et al., 2011). Prior to PCA, all subjective ratings were standardised using z-score normalisation (mean = 0, standard deviation = 1) to ensure comparability across variables. The number of components to retain was determined by inspecting the cumulative explained variance. The first principal component, which accounted for the largest proportion of shared variance across all dimensions, was used as a unidimensional composite score. This score, referred to as the subjective NVB score, served as a summary index of participants' overall impressions of the players' NVB. It was subsequently used in analyses examining the effects of play context on subjective NVB ratings, as well as the relationship between subjective and objective computable measures of NVB. The magnitude of repeated measures correlation was interpreted as small (r_{rm} = 0.1), medium (r_{rm} = 0.3), or large (r_{rm} = 0.5) (Cohen, 2013). Statistical significance was set at $p<0.05$ with 95% confidence intervals.

Results

Objective Computable Measures of NVB in Different Play Contexts

The analysis encompassed 132 data points (i.e. observations of players performing events) and corresponded to 28 individual players. For each player, mean objective NVB features were computed for each event. The findings are summarised in Figs. 5, 6, and 7.

In Fig. 5, the results of the LMMs for the log-transformed total body bounding box volume and upper body convex hull volume were presented. The analysis revealed a significant main effect of play context on both total body bounding box volume $F\,(2,117) = 104.2, p < 0.001$ and upper body convex hull volume $F\,(2,114) = 38.6, p < 0.001$. The effect size of the play context on total body bounding box volume was large ($\omega^2 = 0.63$, $CI\,95\%\,[0.55,1]$)

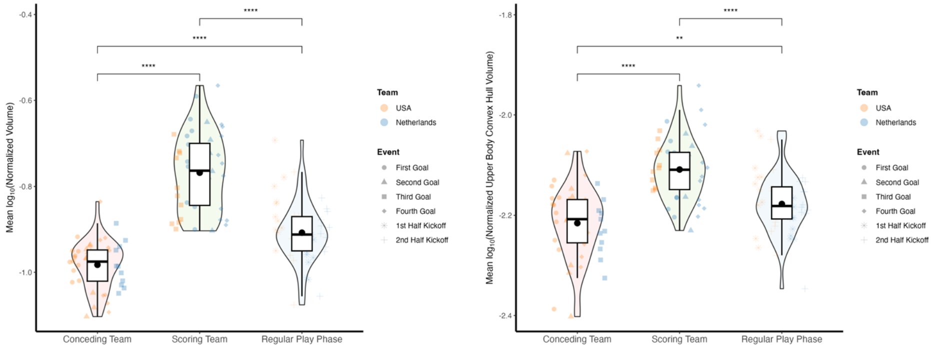

Fig. 5 Boxplots of total body bounding box volume (left) and upper body convex hull volume (right) across play contexts. The boxplots illustrate the distribution of mean values of log-transformed normalised values for each player in each play context. The horizontal lines within the boxes represent the median, while the black dots indicate the mean values for each play context. The asterisks indicate significant differences between play contexts, with * indicating $p<0.05$, ** indicating $p<0.01$, *** indicating $p<0.001$ and **** indicating $p<0.0001$ (Kassambara, 2023)

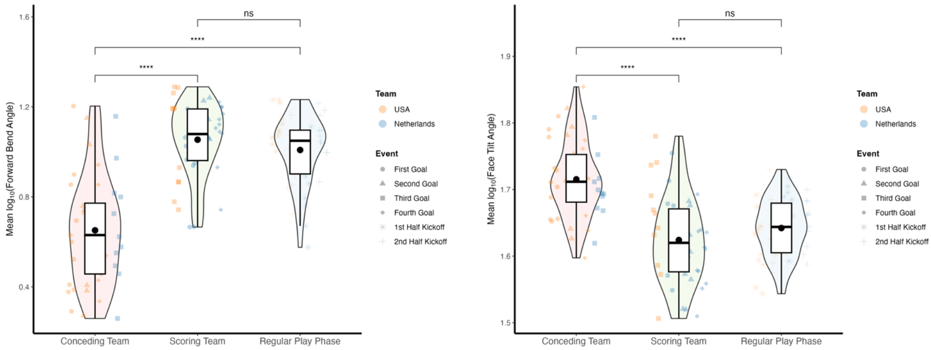

Fig. 6 Boxplots of body bend angle (left) and face tilt angle (right) across play contexts. The boxplots illustrate the distribution of mean values of log-transformed values for each player in each play context. The horizontal lines within the boxes represent the median, while the black dots indicate the mean values for each play context. The asterisks indicate significant differences between play contexts, with * indicating $p<0.05$, ** indicating $p<0.01$, *** indicating $p<0.001$, **** indicating $p<0.0001$ and ns indicating no significant difference (Kassambara, 2023)

and on upper body convex hull volume was large ($\omega^2 = 0.39$, $CI\,95\%\,[0.28, 1]$). The post-hoc pairwise comparisons using Sheffe's method revealed that the total body bounding box volume was significantly smaller in the "Conceding Team" context compared to both the "Scoring Team" context ($estimate = -0.22$, $SE = 0.02$, $t\,(123) = -14.1$, $p < 0.0001$, *Cohen's d* $= -3.2$, $CI\,95\%\,[-3.8, -2.59]$) and the "Regular Play Phase" context ($estimate = -0.08$, $SE = 0.01$, $t\,(109) = -5.17$, $p < 0.0001$, *Cohen's d* $= -1.12$, $CI\,95\%\,[-1.57, -0.67]$). In contrast, the total body bounding box volume was significantly larger in the "Scoring Team" context compared to the "Regular Play Phase" context ($estimate = 0.14$, $SE = 0.01$, $t\,(113) = 9.48$, $p < 0.0001$, *Cohen's d* $= 2.07$, $CI\,95\%\,[1.57, 2.58]$). Additionally, the upper body convex hull volume was significantly smaller in the "Conceding Team" context compared to both the "Scoring Team" con-

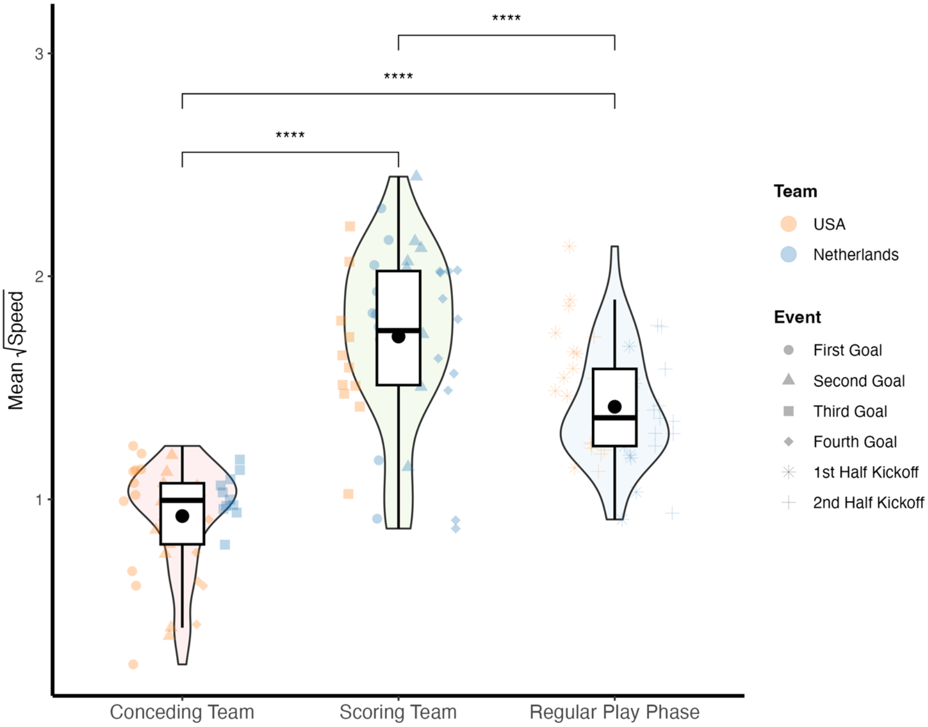

Fig. 7 Boxplot of instantaneous speed across play contexts. The boxplots illustrate the distribution of mean values of square root transformed values for each player in each play context. The horizontal lines within the boxes represent the median, while the black dots indicate the mean values for each play context. The asterisks indicate significant differences between play contexts, with * indicating $p<0.05$, ** indicating $p<0.01$, *** indicating $p<0.001$ and **** indicating $p<0.0001$ (Kassambara, 2023)

text ($estimate = -0.11$, $SE = 0.01$, $t\,(120) = -8.63$, $p < 0.0001$, *Cohen's d* $= -1.98$, $CI\,95\%\,[-2.49, -1.46]$) and the "Regular Play Phase" context ($estimate = -0.04$, $SE = 0.01$, $t\,(108) = -3.37$, $p < 0.0001$, *Cohen's d* $= -0.73$, $CI\,95\%\,[-1.18, -0.29]$). In contrast, the upper body convex hull volume was significantly larger in the "Scoring Team" context compared to the "Regular Play Phase" context ($estimate = 0.07$, $SE = 0.01$, $t\,(112) = 5.65$, $p < 0.0001$, *Cohen's d* $= 1.24$, $CI\,95\%\,[0.78, 1.71]$).

In Fig. 6, the results of the LMMs for the log-transformed body bend angle and face tilt angle were presented. The analysis revealed a significant main effect of play context on both body bend angle $F\,(2, 120) = 62.2, p < 0.001$ and face tilt angle $F\,(2, 107) = 49.1, p < 0.001$. The effect size of the play context on body bend angle was large ($\omega^2 = 0.5$, $CI\,95\%\,[0.39, 1]$) and on face tilt angle was large ($\omega^2 = 0.47$, $CI\,95\%\,[0.35, 1]$). The post-hoc pairwise comparisons using Sheffe's method revealed that the body bend angle was significantly smaller in the "Conceding Team" context compared to both the "Scoring Team" context ($estimate = -0.4$, $SE = 0.04$, $t\,(127) = -9.99$, $p < 0.0001$, *Cohen's d* $= -2.22$, $CI\,95\%\,[-2.74, -1.7]$) and the "Regular Play Phase" context ($estimate = -0.36$, $SE = 0.04$, $t\,(112) = -9.13$, $p < 0.0001$, *Cohen's d* $= -1.97$, $CI\,95\%\,[-2.46, -1.47]$). In contrast, the body bend angle was not significantly different between the "Scoring Team" context and the "Regular Play Phase" context ($estimate = 0.05$, $SE = 0.04$, $t\,(116) = 1.16$,

$p = 0.51$). Additionally, the face tilt angle was significantly larger in the "Conceding Team" context compared to both the "Scoring Team" context ($estimate = 0.09$, $SE = 0.01$, $t\,(111) = 9.08$, $p < 0.0001$, *Cohen's d* = 2.13, $CI\,95\%\,[1.59, 2.67]$) and the "Regular Play Phase" context ($estimate = 0.07$, $SE = 0.01$, $t\,(105) = 8.03$, $p < 0.0001$, *Cohen's d* = 1.76, $CI\,95\%\,[1.26, 2.25]$). In contrast, the face tilt angle was not significantly different between the "Scoring Team" context and the "Regular Play Phase" context ($estimate = -0.02$, $SE = 0.01$, $t\,(107) = -1.66$, $p = 0.26$).

In Fig. 7, the results of the LMMs for the square root transformed instantaneous speed were presented. The analysis revealed a significant main effect of play context on instantaneous speed $F\,(2, 117) = 90.5$, $p < 0.001$. The effect size of the play context on instantaneous speed was large ($\omega^2 = 0.6$, $CI\,95\%\,[0.51, 1]$). The post-hoc pairwise comparisons using Sheffe's method revealed that the instantaneous speed was significantly smaller in the "Conceding Team" context compared to both the "Scoring Team" context ($estimate = -0.8$, $SE = 0.06$, $t\,(122) = -13.2$, $p < 0.0001$, *Cohen's d* = -3.01, $CI\,95\%\,[-3.59, -2.42]$) and the "Regular Play Phase" context ($estimate = -0.49$, $SE = 0.06$, $t\,(109) = -8.47$, $p < 0.0001$, *Cohen's d* = -1.84, $CI\,95\%\,[-2.33, -1.35]$). In contrast, the instantaneous speed was significantly larger in the "Scoring Team" context compared to the "Regular Play Phase" context ($estimate = 0.31$, $SE = 0.06$, $t\,(112) = 5.32$, $p < 0.0001$, *Cohen's d* = 1.17, $CI\,95\%\,[0.71, 1.63]$).

Subjective Ratings of NVB

Subjective ratings of players' NVB were obtained from an average of 30 participants for each video clip (total N=496; 248 per video type: 3D pose or tactical). To ensure robust analysis, these ratings were aggregated by calculating the mean for each player, event, and video type. Repeated measures correlation analyses were then performed across the four rating dimensions within each video type (eight observed variables), yielding a mean repeated measures correlation of $r_{rm} = 0.91 \pm 0.05$. This high level of intercorrelation indicates strong consistency among the subjective rating dimensions (see Supplementary Material: 'Subjective Ratings of Nonverbal Behaviour'). Given this strong association, principal component analysis (PCA) was conducted to identify a common underlying factor.

The PCA results showed that the first principal component explained approximately 92% of the total variance, with nearly equal loadings (0.35 or 0.36) across all rating dimensions. This suggests that the first principal component represents a general evaluative dimension of players' NVB, reflecting overall nonverbal presence and expressiveness. The second principal component accounted for only about 4% of the variance, indicating minimal additional contribution. Based on these findings, the subjective ratings were projected onto the first principal component to derive a single "subjective NVB score," which served as a comprehensive summary of participants' impressions of each player's NVB (see Supplementary Material: 'Subjective Ratings of Nonverbal Behaviour').

In Fig. 8, the results of the LMMs for the subjective NVB score were presented. The analysis revealed a significant main effect of play context on subjective NVB score $F\,(2,\ 115) = 231.9$, $p < 0.001$. The effect size of the play context on subjective NVB score was large ($\omega^2 = 0.8$, $CI\,95\%\,[0.74,\ 1]$). The post-hoc pairwise comparisons using Sheffe's method revealed that the subjective NVB score was significantly lower in the "Conceding Team" context compared to both the "Scoring Team" context ($estimate = -6.39$,

$SE = 0.3, t\,(123) = -20.96, p < 0.0001, Cohen's\ d = -4.71, CI\,95\%\,[-5.19, -4.23]$) and the "Regular Play Phase" context ($estimate = -3.93$, $SE = 0.3$, $t\,(108) = -13.25$, $p < 0.0001$, $Cohen's\ d = -2.9$, $CI\,95\%\,[-3.33, -2.46]$). In contrast, the subjective NVB score was significantly higher in the "Scoring Team" context compared to the "Regular Play Phase" context ($estimate = 2.46$, $SE = 0.3$, $t\,(112) = 8.17$, $p < 0.0001$, $Cohen's\ d = 1.81$, $CI\,95\%\,[1.31, 2.31]$).

Repeated Measures Correlation between Objective Measures and Subjective NVB Scores

In order to investigate the relationship between objective computable measures of NVB and subjective NVB scores, a series of repeated measures correlation analyses were conducted. The results of these analyses are presented in Fig. 9. The findings indicate a significant correlation with large effect sizes between objective computable measures of NVB and subjective NVB scores. Specifically, the total body bounding box volume, upper body convex hull volume, body bend angle and instantaneous speed exhibited significant positive correlations with the subjective NVB scores, while the face tilt angle exhibited a significant negative correlation.

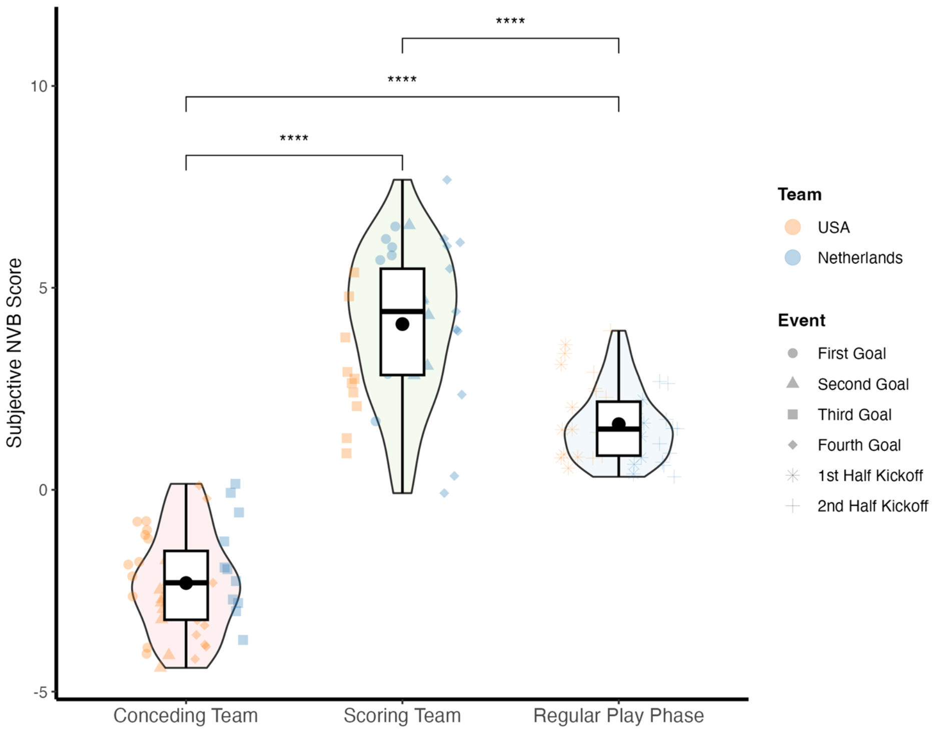

Fig. 8 Boxplot of subjective NVB scores across play contexts. The boxplots illustrate the distribution of mean values of subjective NVB scores for each player in each play context. The horizontal lines within the boxes represent the median, while the black dots indicate the mean values for each play context. The asterisks indicate significant differences between play contexts, with * indicating $p<0.05$, ** indicating $p<0.01$, *** indicating $p<0.001$ and **** indicating $p<0.0001$ (Kassambara, 2023)

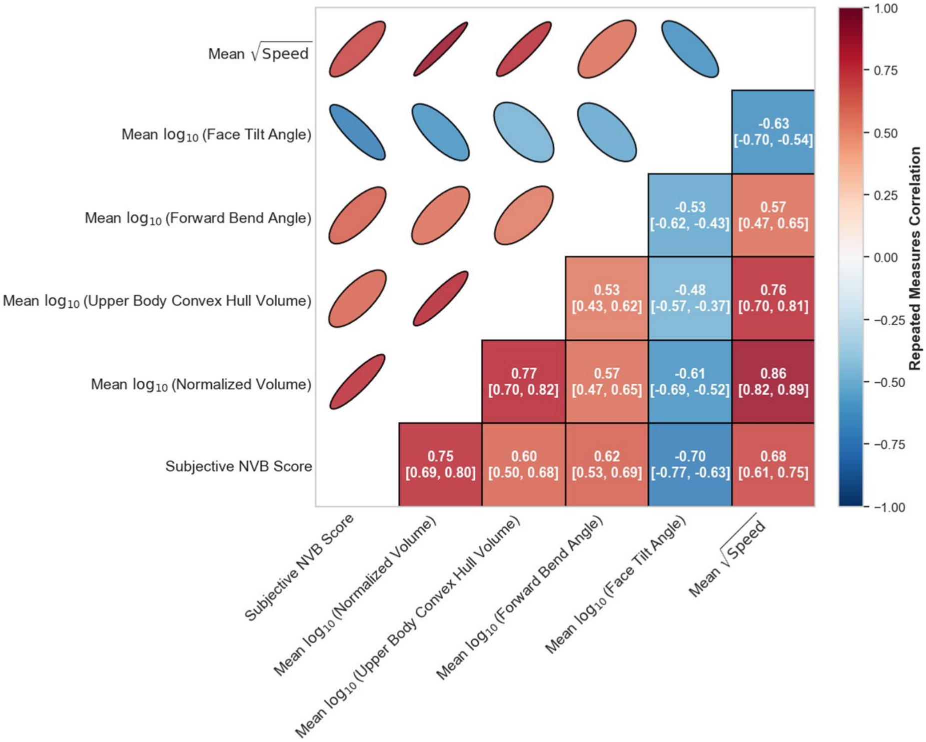

Fig. 9 Repeated measures correlation between objective computable NVB measures and subjective NVB scores. The lower triangle of the matrix displays the repeated measures correlation coefficients with 95% confidence intervals, while the upper triangle displays and visualizes the same correlations using ellipses whose orientation and eccentricity reflect the strength and direction of the association

Discussion

The objective of the present study was to identify and validate objective, computable indicators derived from 3D pose data that can quantitatively represent aspects of NVB, and to examine how these indicators correspond to subjective human impressions and evaluations.

The initial findings of this study reveal clear and significant differences in objective computable NVB indicators across different play contexts. Specifically, players on the conceding team exhibited smaller total body bounding box volumes and upper body convex hull volumes compared to players on the scoring team. This pattern suggests that conceding players tend to adopt more contracted and less expansive postures, which may reflect reduced confidence or a defensive psychological state. These results are consistent with previous research demonstrating that expansive body language is associated with positive internal states, such as confidence and dominance, while contracted postures are linked to lower status or defensive intentions (Carney et al., 2005, 2010; Furley & Roth, 2021). Additionally, a significant difference was observed in face tilt angle between the two groups. This finding is consistent with previous studies suggesting that increased face tilt is a characteristic nonverbal expression of shame or embarrassment (Keltner, 1995; Tracy & Matsumoto, 2008). In this context, the larger face tilt angle among conceding players suggests a tendency toward more defensive or withdrawn expressions following negative match events. Furthermore, analysis of instantaneous speed revealed that players on the conceding team moved significantly more slowly than those on the scoring team or in the Regular Play Phase. This reduced movement speed may indicate a more passive or reactive response to conceding a goal, whereas increased speed among scoring players may reflect greater assertiveness and confidence. These findings support the notion that NVB, including movement dynamics, serves as an important channel for expressing emotional states and intentions (Gross et al., 2012; Montepare et al., 1987).

A notable finding of this study was that the body bend angle of players on the conceding team was significantly smaller than that of both the scoring team and during the Regular Play Phase. This outcome was unanticipated, given that earlier studies have predominantly correlated forward-leaning posture with feelings of shame or embarrassment (Keltner, 1995; Tracy & Matsumoto, 2008). Contrary to these established findings, the present results indicate that players on the conceding team adopted a more upright posture than anticipated. This divergence highlights the complexity of NVB in competitive sports and underscores the need for further research to clarify the nuanced relationship between body posture and psychological states in these contexts.

Regarding subjective ratings, human observers consistently rated the NVB of players on the conceding team lower than those on the scoring team, both during the Regular Play Phase and following scoring events. This pattern aligns with previous studies by Furley and Schweizer (2013) and Furley and Schweizer (2015), which demonstrated that observers systematically perceive players on the trailing (conceding) side as less dominant, less proud, and less confident than those on the leading (scoring) side. Furthermore, observers' subjective NVB scores are closely and consistently aligned with objective, computable measures of NVB. It is conceivable that observers may employ these cues to inform inferences regarding players' psychological states and intentions (i.e., cue utilisation). The close correspondence between computationally derived NVB features and human judgments has the potential to support the ecological validity of the selected objective metrics. Furthermore, it has the potential to emphasise the interpretability and relevance of these metrics for capturing expressive behaviours that are meaningful to human perceivers.

While this study provides valuable insights into the relationship between objective NVB metrics and subjective human judgments in football, several limitations should be considered. First, the generalizability of the findings is restricted by the use of data from a single match involving only male professional players. The unique characteristics of this match, including the teams and specific situational context, may have influenced the observed NVB patterns. Second, the analysis was limited to a small set of pre-defined events (goals and the start/end of halves). Although these moments are highly relevant, other important events in football—such as fouls, near misses, saves, and substitutions—were not included and may also elicit meaningful NVB. Third, 3D pose estimation from video can be affected by inaccuracies, particularly in cases of player occlusion, which may introduce noise into the calculation of objective features. Finally, the present study focused solely on bodily expressions and did not incorporate facial information. In authentic competitive scenarios, facial and bodily expressions can exhibit dynamic congruence or incongruence, thereby conveying nuanced meanings to both participants and observers. Consequently, it is imperative to acknowledge that the objective computational features selected are capable of capturing only a limited proportion of the intricate information inherent. Despite these limitations, this study highlights several promising directions for future research. Integrating whole-body NVB data with other modalities, such as facial expressions, could provide a more comprehensive understanding of nonverbal communication. Advanced machine learning techniques may enable predictive modeling to determine whether computable NVB features can forecast performance, tactical decisions, or match outcomes.

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Author Contributions A.U. was responsible for the design of the data processing pipeline and the conduction of the statistical analyses, in addition to the drafting of the manuscript. P.F. and D.M. were responsible for the conceptualisation of the study, the review and editing of the manuscript, and the management and supervision of the project.

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Data Availability The 3D pose data is the intellectual property of FIFA and is not accessible to the public. Consequently, the authors are unable to share it. Notwithstanding, the data and code that were utilised for the relevant analysis have been uploaded to the Open Science Framework (at https://osf.io/s36h5/?view_only=f402f9c014e84b87ac807b1c9f4efc9d).

Declarations

Competing interests The authors declare no competing interests.

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